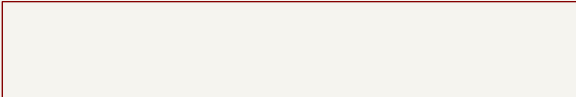
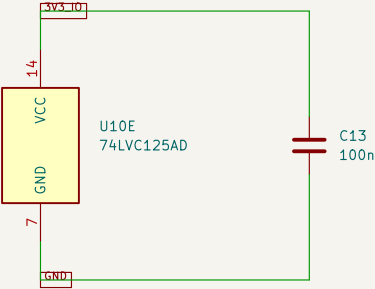
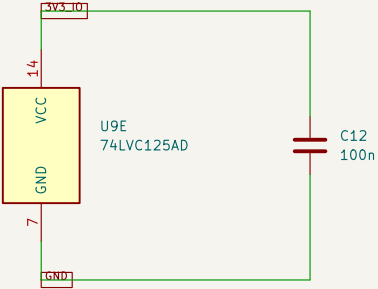
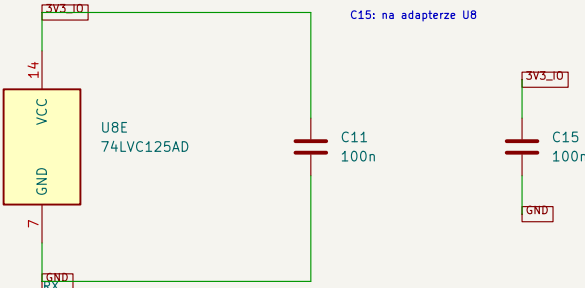
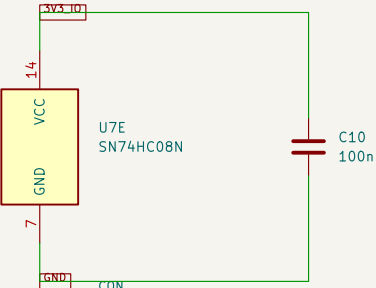
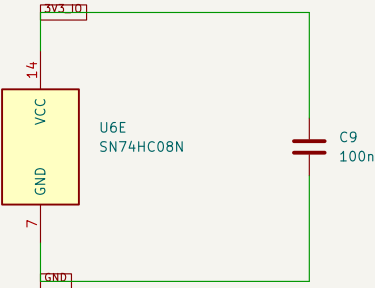
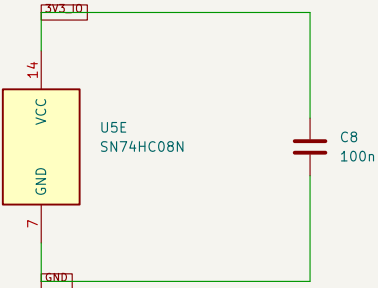
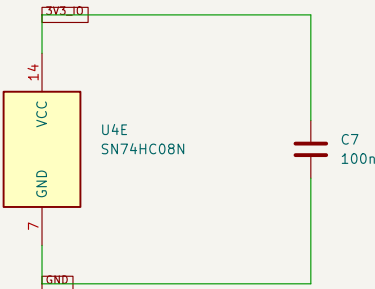
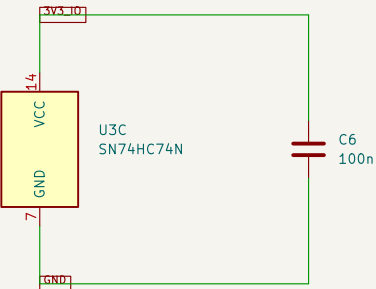
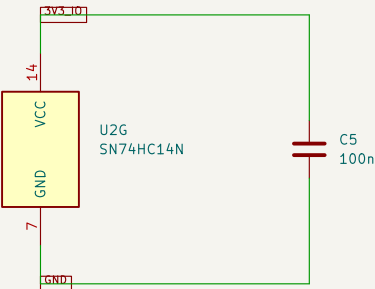
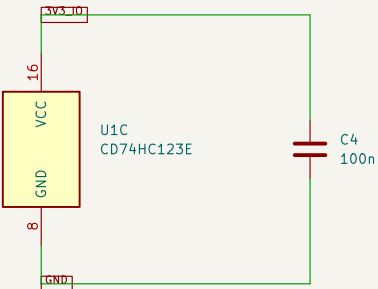


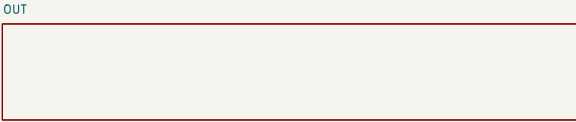
Zasilanie 3V3_IO z P02. 5V_SYS z LV04 jest dostępne na J1.1 i TP13; nie zasila logiki.



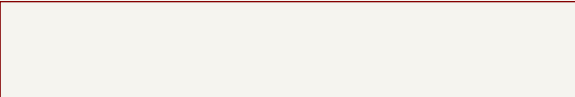
File: CON.kicad_sch



File: ILK.kicad_sch



File: OUT.kicad_sch



File: RX.kicad_sch



File: WD.kicad_sch

U11: MCP100 w bondout D, 1 RESET / 2 VDD / 3 GND. Lokalny reset nie jest podłączony push-pull do SAFE_N.

Hardware interlock, watchdog and manual arm; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /

File: P04.kicad_sch

Title: EGRLab P04 SAFE

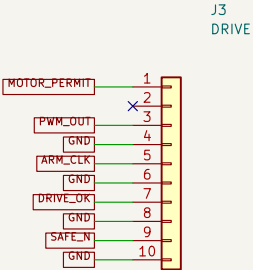
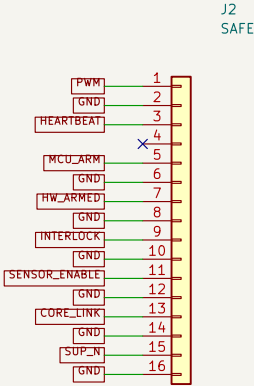
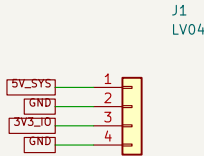
Size: A3 Date: 2026-09-28

Rev: P04-R2.2

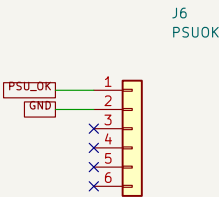
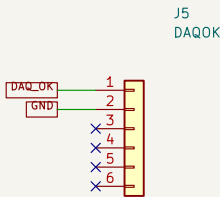
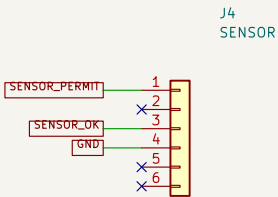
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Id: 1/6

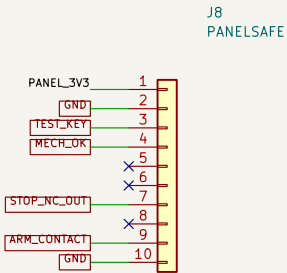
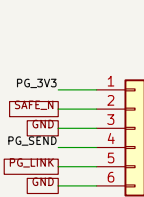
J1 LV04: przewody AWG22 200 mm. J2 SAFE: tasma 16 zyl AWG28 150 mm. PTH + kotwa 12 mm.



Klucze: SAFE 4 / DRIVE 2 / SENSOR 2 / DAQOK 4 / PSUOK 5. Pad klucza NC; usuniety styk w gniezdzie.



SENSOR M2.2: IDC 6p; 1 permit, 2 KEY, 3 ready, 4 GND, 5/6 NC. Wymagana zgodna wiazka P08.



PANELSAFE: STOP NC miedzy 1 i 7. ARM NO miedzy 9 i 10. Nie laczyc szyny P00 3V3 z P04 3V3_IO.

R2: 3V3 wychodzi z plytki tylko przez rezystory: J8.1 R40 100R, J7.1 R39 1k, petla PG R38 1k (zwarcie w wiazce nie kladzie 3V3_IO).

Hardware interlock, watchdog and manual arm; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /CON/

File: CON.kicad_sch

Title: EGRLab P04 SAFE

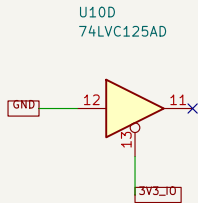
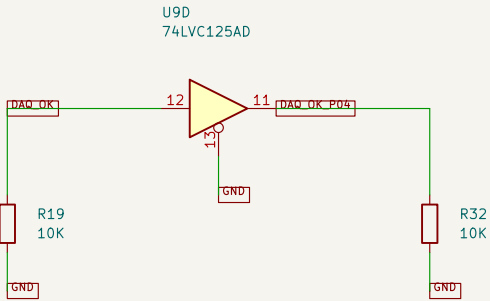
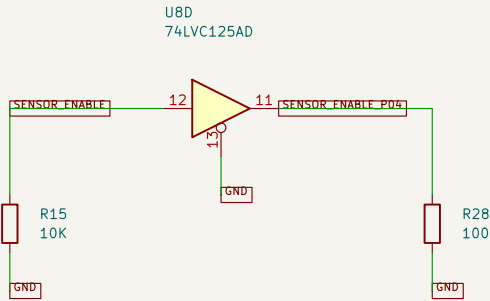
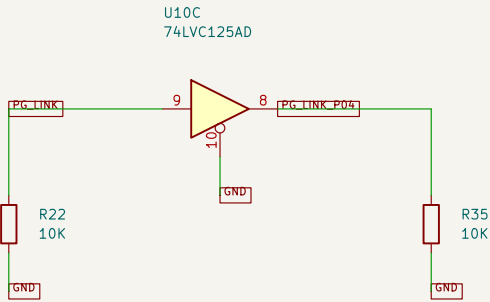
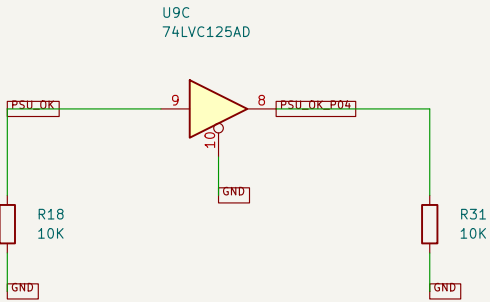
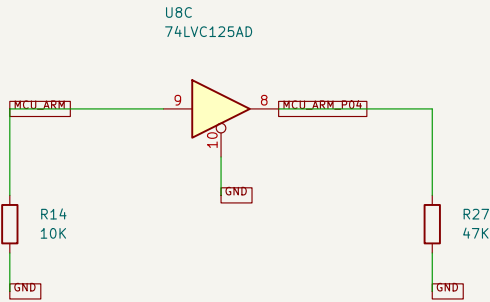
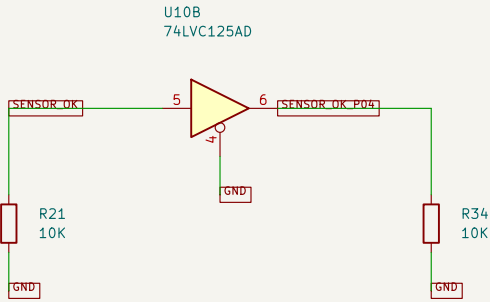
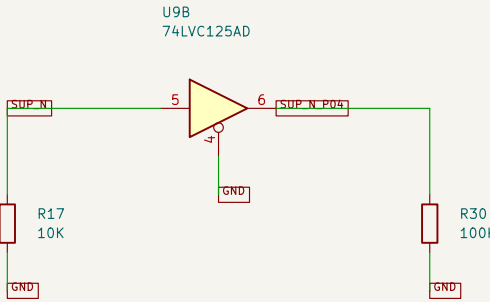
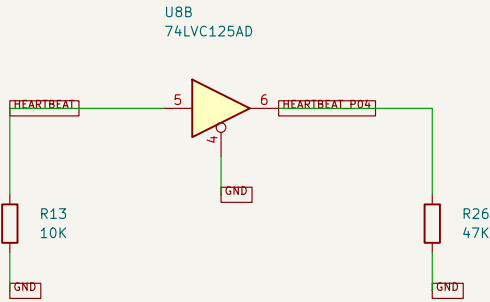
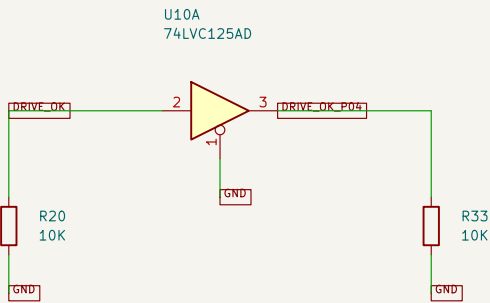
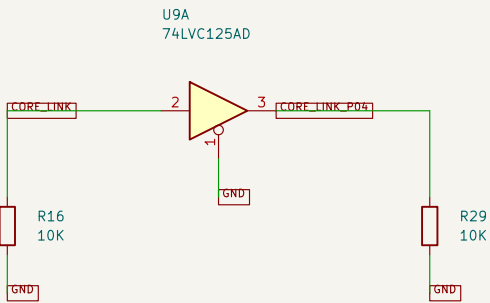
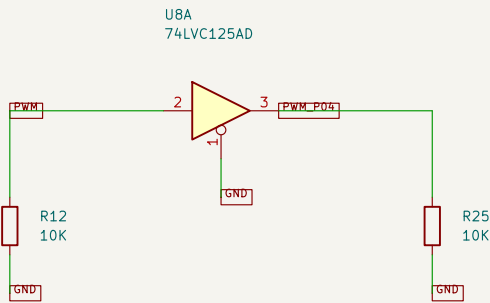
Size: A3

Date: 2026-09-28

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74LVC125AD Nexperia: loff; pin wejścia nie zasila nieaktywnej domeny. Nie zastąpić układem HC125.

Hardware interlock, watchdog and manual arm; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /RX/

File: RX.kicad_sch

Title: EGRLab P04 SAFE

Size: A3

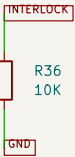
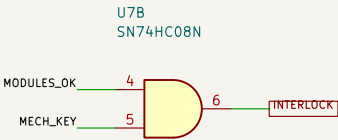
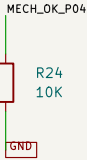
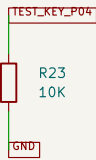
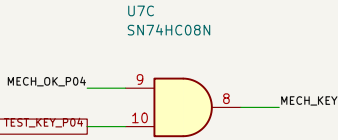
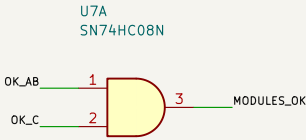
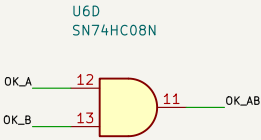
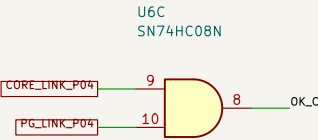
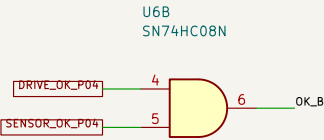
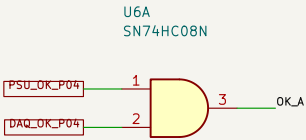
Date: 2026-09-28

Rev: P04-R2.2

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Id: 3/6

MODULES_OK = PSU_OK & DAQ_OK & DRIVE_OK & SENSOR_OK & CORE_LINK & PG_LINK (po buforach).



R2: TEST_KEY i MECH_OK z panelu przez R41/R42 1k; pulldowny R23/R24 po stronie bramek (przerwa w R41/R42 = L).

INTERLOCK = MODULES_OK & MECH_OK & TEST_KEY. P11 MECH_OK zawiera klucz, oba styki LOGGER i petle TEST.

READY nie zależy od przełącznika, który ma uzbroić. Powrót READY nie wywołuje nowego zbocza ARM.

Hardware interlock, watchdog and manual arm; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /ILK/
File: ILK.kicad_sch

Title: EGRLab P04 SAFE

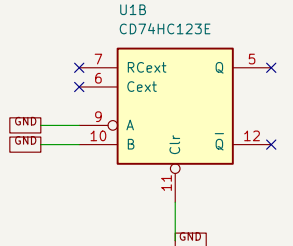
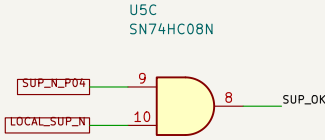
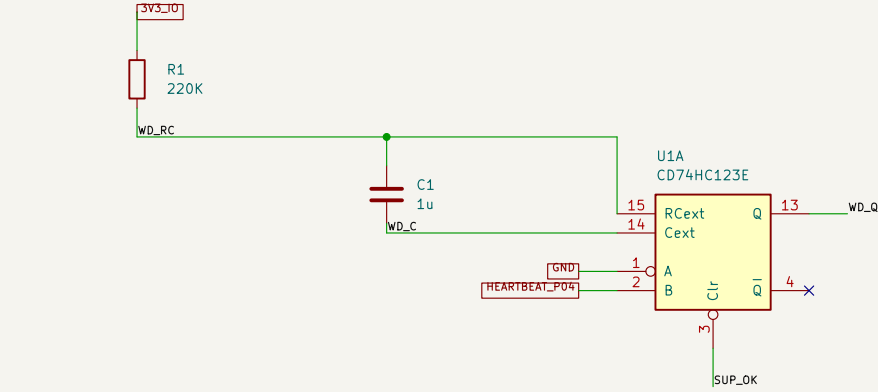
Size: A3
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Date: 2026-09-28

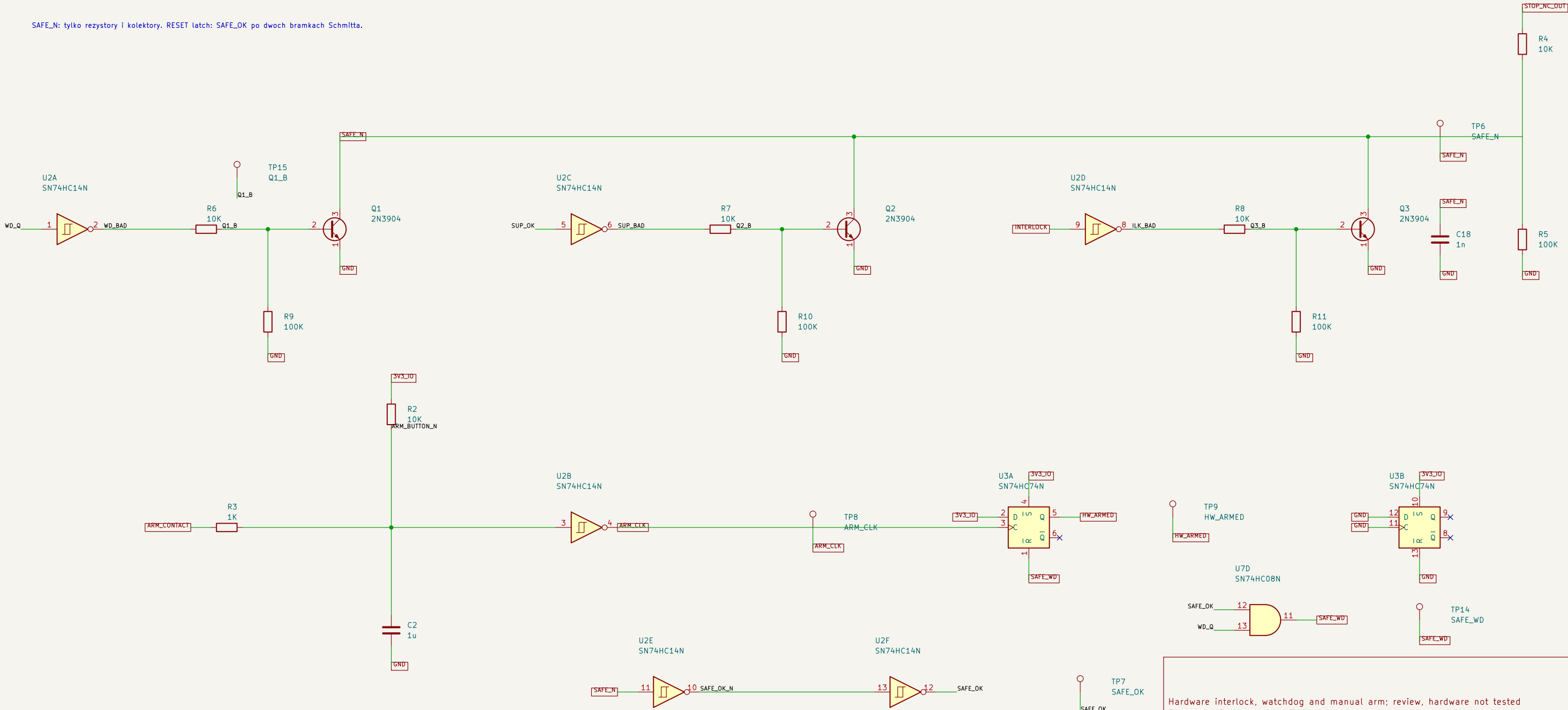
Rev: P04-R2.2

Id: 4/6

U1 CLR = SUP_OK, niezależnie od SAFE_N. C1 pomiędzy pinami 15 i 14; pin 14 NIE jest masa.



SAFE_N: tylko rezystory i kolektory. RESET latch: SAFE_OK po dwóch bramkach Schmitta.



R2: SAFE_WD = SAFE_OK & WD_Q (U7D) kasuje zatrask U3 i obie zgody także przy uszkodzonym Q1. C18 1n COG na SAFE_N.

WD: cel 50-150 ms przy 3V3, do pomiaru. ARM: zwolnic i nacisnac ponownie po bledzie.

Hardware interlock, watchdog and manual arm; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /WD/

File: WD.kicad_sch

Title: EGRLab P04 SAFE

Size: A3

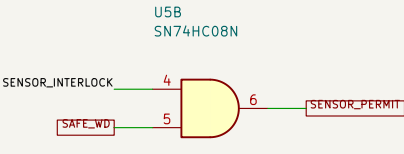
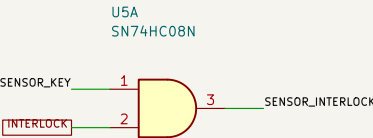
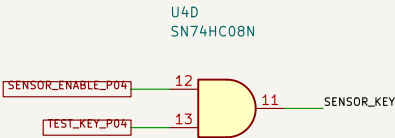
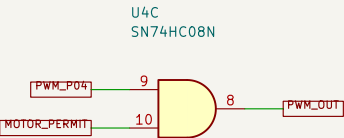
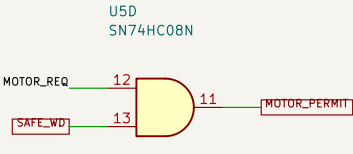
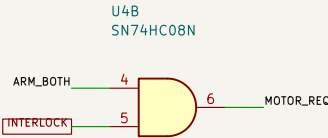
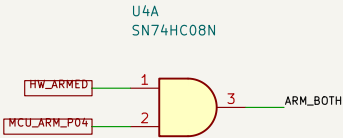
Date: 2026-09-28

Rev: P04-R2.2

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Id: 5/6

MOTOR_PERMIT = HW_ARMED & MCU_ARM & INTERLOCK & SAFE_WD. PWM_OUT = PWM & MOTOR_PERMIT. SAFE_WD = SAFE_OK & WD_Q.



SENSOR_PERMIT = SENSOR_ENABLE & TEST_KEY & INTERLOCK & SAFE_WD. Fizyczny ARM nie jest potrzebny do samego czujnika.

P07 pozostaje HOLD do kontroli rzeczywistego BTS7960. P04 nie ustala prądu ani kierunku mostka.

Hardware interlock, watchdog and manual arm; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /OUT/

File: OUT.kicad_sch

Title: EGRLab P04 SAFE

Size: A3

Date: 2026-09-28

Rev: P04-R2.2

KiCad E.D.A. 10.0.6

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